CS403P - Quiz No.3

Date: 02 January 2026

Q.1 If relation M has C tuples and relation N has D tuples, then their Cartesian product will have _____ tuples.

A. C∩D

B. C^D

C. CUD

D. C*D

Q.2 A relation is in second normal form if and only if _____ and all the non key attributes are fully functionally dependent on the key.

A. It has multivalued attributes

B. It is in first normal form

C. It is not in first normal form

D. It has inconsistent data

Q.3 Suppose a relation named EMPLOYEE contains attributes Emp_name, Emp_designation, Emp_address and Emp_salary. Which of the following query will be used to SELECT all those employees whose designation is Admin officer?

A. ϕ designation = 'Admin officer'(EMPLOYEE)

B. α designation = 'Admin officer' (EMPLOYEE)

C. σ designation = 'Admin officer' (EMPLOYEE)

D. designation = 'Admin officer' (EMPLOYEE)

Q.4 Which of the following are unary operators?

A. UNION and PROJECT

B. SELECT and CARTESIAN

C. SELECT and PROJECT

D. PROJECT and CARTESIAN

Q.5 2nd Normal Form is based on which of the following dependency?

A. Reflexive dependency

B. Transitive dependency

C. Partial dependency

D. Full functional dependency

Q.6 If there are two relations X and Y then the difference of these two relations (i.e. X-Y) will be set of tuples that _____.

A. Appear in both X and Y

B. Appear in Y but not in X

C. Do not appear in both X and Y

D. Appear in X but do not appear in Y

Q.7 Consider the following functional dependency: A → B In above functional dependency, attribute B is_____ on A.

A. Reflected

B. Terminated

C. Dependent

D. Projected

Q.8 The only time we are concerned about 2nd Normal Form is when _____.

A. Primary key consists of single attribute

B. Primary key is composite

C. Primary key contains redundant values

D. Primary key contains non-unique values

Q.9 Suppose A → B, then which of the following is valid Functional dependency for augmentation?

A. AD → B

B. AC → BC

C. A → C

D. AB → C

Q.10 A table is in the 2nd Normal Form if there is no_____.

A. Functional dependency

B. Transitive dependency

C. Reflexive dependency

D. Partial dependency
